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| **Type: DSC2-6(Minor)**<br>**B.Sc(ECS)-II ( Semester IV )**<br>**Course Title: Software Testing and Quality Assurance**<br>**(Paper Code: )** |
| **Credits: Theory – (2)** **Practical – (1)** |
| **Total Lectures:** 30 Hrs. **Contact Hrs. (L) :** 2 |
| **University Evaluation:** 30 Marks **Internal Evaluation:** 20 Marks |

**Course Outcomes:**

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| 1. | List a range of different software testing techniques and strategies and be able to apply specific (automated) unit-testing method for the projects. |
| 2. | Distinguish characteristics of structural testing methods. |
| 3. | Demonstrate the integration testing which aims to uncover interaction and compatibility problems as early as possible. |
| 4. | Discuss about the functional and system testing methods. |
| 5. | Demonstrate various issues for object-oriented testing. |

| Unit | Content | Lect. |
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| **I** | **Introduction To Software Testing:**<br>Importance or need of software testing<br>Differences between Manual and Automation Testing<br>**Introduction to White Box Testing:**<br>Advantages and Disadvantages of White box testing<br>Static Techniques: Informal Reviews, Walkthroughs, Technical Reviews, Inspection<br>Dynamic Techniques or Structural Techniques<br>Statement Coverage Testing, Branch Coverage Testing<br>Path Coverage Testing, Conditional Coverage Testing, Loop Coverage Testing<br>**Introduction to Black Box Testing:**<br>Advantages and Disadvantages of black box testing<br>Black Box Techniques: Boundary Value Analysis, Equivalence Class Partition, State Transition Technique, Cause Effective Graph, Decision Table, Use Case Testing<br>Experienced Based Techniques: Error guessing, Exploratory testing Levels of Testing<br>Functional Testing<br>Integration Testing and types - Top Down , Bottom Up , Non Incremental<br>System Testing<br>Acceptance Testing- Alpha and Beta<br>Smoke Testing | **15** |

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| | Regression Tesng- Unit , Regional, Full Non Functional Testing<br>Adhoc Testing<br>Performance Testing: Load Testing, Stress Testing, Volume Testing, Soak Testing<br>Recovery Testing | |
| **II** | **Test cases design Techniques:**<br>Introduction to Test Case and Types.<br>Test Case Template<br>How to write a test case and examples<br>Preparing Review Report Software Test Life cycle<br>Software Test Life Cycle:<br>Writing Test Plan<br>Preparing Traceability Matrix<br>Writing Test Execution Report and Summary Report<br>**Defect Life Cycle:**<br>Bug/ Defect Life Cycle: Difference between Bug, Defect, Failure, Error<br>Defect Tracking and Reporting<br>Types of Bugs, Identifying the Bugs, Reporting the Bugs<br>**Introduction to automated testing**- Install and configure selenium testing tool, Case study: Design test case for login page, Internet Banking Login, and Online shopping.<br>**TestNG Testing framework:**<br>Introduction TestNG, Annotation, Methods, Create Auto Testcase, create auto report. | **15** |


| **Reference Books:** | |
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| 1. | The art of Software Testing–Glenford J. Myers |
| 2. | Lessons learned in Software Testing– Cem Kaner, James Bach, Bret Pettichord |
| 3. | A Practitioner’s Guide to Software Test Design- Lee Copeland |